AODet: Aerial Object Detection Using Transformers for Foreground Regions

Xiaoming Wang, Hao Chen, Xiangxiang Chu, and Peng Wang

***Abstract*— Aerial object detection is an important task and has received significant attention in recent years. Aerial images typically depict small and sparse instances against a simple background. Nevertheless, the simple background can only provide limited information. Based on the observation, we present a new transformer-based framework for aerial object detection. In contrast to previous methods that address sparsity through multistage pipelines involving region-of-interest (RoI) techniques or sparse convolutions, our method, referred as AODet, enjoy two significant advantages: 1) AODet is a simple yet accurate object detector which is specialized for aerial object detection. AODet identifies the background regions earlier and then only operates on the regions which most likely include the foreground objects, thereby significantly reducing the redundant computations. The utilization of transformer exploits more context information between foreground regions, helping to retain high-quality detection results and 2) instead of involving the sparse operations like sparse convolutions or clustering algorithms/RoI operations, AODet employs transformer to detect objects from foreground proposals. Our approach is simpler and can be easily implemented with simple tensor manipulations. Extensive experiments have conducted on VisDrone and DOTA. AODet achieves 40.9 AP on Visdrone and 79.6 mAP DOTA, demonstrating the effectiveness of AODet.**

***Index Terms*— Aerial object detection, transformer.**

I. Introduction

In RECENT years, aerial object detection has garnered significant attention from researchers, primarily owing to its large number of practical applications. With the great progress made in general object detection, aerial object detection, as a specific branch of object detection, also developed fast in efficiency and effectiveness. However, there are some differences between aerial object detection and general object detection. First of all, aerial images are typically captured in a high view from ground, attributing to small size of objects. For instance, it is usually 12 to 20 pixel in VisDrone. While in MS COCO, it is usually more than 32 pixels. Second, the objects are usually very sparse. Despite the specific scenes like parking lot, most of the scenes have sparse objects, which means the useful instances only occupy a small fraction of the image. As illustrated in Fig. 1, background pixels are up to 90.3% on Visdrone and 86.6% on DOTA, which are in sharp contrast to the general object detection benchmark MS COCO where only 56.0% pixels are background. Third, most of the background regions are very simple (e.g., sky and road) and can only provide limited information.

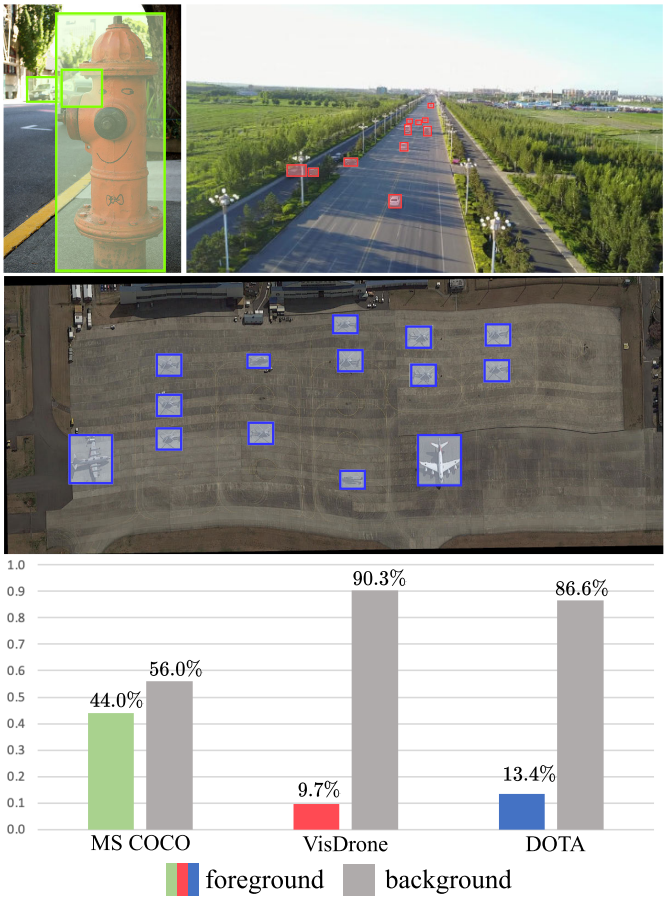


Fig. 1. Ratios of foreground and background pixels on MS COCO, VisDrone, and DOTA, respectively.

Based on the above observation, it comes to an direct idea that we can ignore the background regions earlier and only focus on the foreground regions to reduce the redundant computations. In this way, the subsequent computation of the model can only operate on the regions where objects are most likely to appear, which can be expected to significantly reduce the inference time.

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Xiaoming Wang is with the School of Computer Science, Ningbo Institute, Northwestern Polytechnical University, Xi'an 710072, China, and also with Meituan, Beijing 100102, China (e-mail: xiaomingwang80@163.com).

Hao Chen is with the College of Computer Science and Technology, Zhejiang University, Hangzhou 310058, China.

Xiangxiang Chu is with Meituan, Beijing 100102, China.

Peng Wang is with the School of Computer Science, Ningbo Institute, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: peng.wang@nwpu.edu.cn).

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Previous methods also explored this aspects. ClusDet [1] extracts candidate regions of objects which contain at least three objects with a relatively lightweight region proposal network (RPN). Subsequently, the second-stage network is applied to the features of these cluster chips to obtain the final detection results. UFPMP-Det [2] proposes unified foreground packing (UFP) module, which predicts subareas that are assembled into a unified mosaic to improve the foreground ratio. Following this, multiproxy (MP) detection network takes the unified images as input and generates predicted bounding boxes. Both methods have been applied to aerial object detection on Visdrone and have achieved impressive performance.

Despite being effective, it is still an open question due to the following issues.

1) Previous methods need complex clustering algorithms, such as iterative cluster merging in ClusDet and foreground region generation in UFPMP-Det. These processes need repeat several times to determine the final subareas and are independent of the neural networks, hampering an end-to-end trainable framework.
2) ClusDet makes use of the region-of-interest (RoI) operations to obtain the cluster regions, while UFPMP-Det employs PHSPPOG method [3] to merge subregions into a unified mosaic. Both approaches require extra effort to implement and is not natively supported by modern deep learning frameworks such as PyTorch [4].
3) Each subregion is independently processed by a detection network without any kind of communication, precluding the possibility of sharing the computations between regions. Consequently, this approach eliminates the chance that each individual region utilize other regions as its context.

Recently, QueryDet [5] proposes to employ sparse convolutions [6] for this task. To be specific, QueryDet first predicts the coarse locations of objects (i.e., foreground regions) on low-resolution features and then refines the detection results on high-resolution features using sparse convolutions. Although sparse convolution allows QueryDet to focus solely on the foreground regions, it still fails to fully utilize contextual information as it does not facilitate communication between disconnected regions. Moreover, sparse convolution is not readily available in modern deep learning frameworks and is challenging to implement and use. Its discontinuous memory layout, stemming from sparsity, also poses a challenge for engineering optimization.

In this article, we bypass the aforementioned issues by taking advantage of the transformer architectures [7] and propose a transformer-based aerial object detector AODet which leverages the above observation in aerial images. One of the most crucial properties of transformers is that its core operation, self-attention, does not make any assumptions about the input format. Thus, the input is not necessarily in the 2-D grid form. This flexibility renders it highly suitable for aerial object detection. Consequently, in our method, all the features from the foreground regions can be viewed as a bag[1] of features. Positional encodings added into the features convey the necessary spatial locations of the features. In this way, AODet does not involve any sparse operations and all the operations can be performed with dense tensor manipulations. Moreover, unlike ClusDet and QueryDet, AODet can take more context information. This is because the self-attention in our method operates on all the foreground features simultaneously, allowing each foreground proposal to access information from any other foreground proposals. As a result, AODet attains an better speed-accuracy trade-off than previous detectors specialized for aerial images.

Here, we summarize our main contributions as follows.

1) We propose AODet, a simple yet accurate object detector specialized for aerial object detection using transformers. AODet identifies the background regions earlier and then only operates on the regions which most likely include the foreground objects, significantly reducing the redundant computations on the background regions. By leveraging the power of transformer-based architecture, AODet can exploit more context information between foreground regions, helping to retain high-quality detection results.
2) Unlike the previous methods, AODet does not involve the sparse operations like Sparse Convolutions or Clustering algorithms/ROI operations. These make our method significantly simpler and can be easily implemented with simple tensor manipulations that are readily available in mainstream deep learning frameworks.

II. Related Work

A. General Object Detection

Object detection mainly consists of two genres, one-stage and two-stage. There are myriad of two-stage detectors like [8], [9], [10], [11], [12], which typically generate a set of RoIs first and classify each of them in the second stage. Specifically, R-CNN [8] yields RoIs using selective search [13]. Each region is then cropped from the original image and then processed by a CNN, which introduces redundant computation. Fast R-CNN [9] enhances R-CNN via an RoI pooling layer which extracts each region from the feature maps. Both rely on separate proposal algorithms and cannot be trained in an end-to-end fashion. Faster-RCNN [10] further adopts a RPN to circumvent this issue, which generates proposals from a set of anchor boxes. The detector can thus be trained in an end-to-end fashion and becomes more efficient. Two stage methods require the bounding box of proposals from the first stage and then align to features in the second stage using RoIPooling [9] or RoIAlign [14]. In contrast, AODet is much simpler by selecting possible foreground pixels of feature maps, which will be directly sent to Transformer detection head for predicting objects.

One-stage detectors make a further simplification, which remove the RoI pooling step and detects objects by a single network. Hence, they generally come with computational efficiency and become popular nowadays. For instance, YOLOv3 [15] predicts bounding-box coordinates directly from an image. Single shot multibox detector (SSD) [16] predicts anchor boxes densely over feature maps at different scales and then performs classification and regression on each target

[1]We use the term "bag" here as the transformer itself is essentially order-agnostic (if positional encodings are not used).

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box. CornerNet [17] detects an object as a pair of key points to depict the top-left and bottom-right corners of a bounding box. The network also predicts an embedding vector to match the key points of the same target. A fully convolutional one-stage object detector (FCOS) [18] reformulates object detection as an anchor-free per-pixel prediction task and invents a "centerness branch" to enhance the overall performance.

B. Aerial Object Detection

Aerial object detection has been explored from different aspects. Some works combine multilevel feature maps to improve the accuracy. CFANet [19] utilizes a cross-layer feature aggregation module and layered associative spatial pyramid pooling module to combine multilevel features, which contribute to better combination of spatial and semantic information while keeping the details of each level of features. FENet [20] extracts additional spatial context to enhance information about small objects. It employs multigranular deformable convolution network and high-resolution feature pyramid network (FPN) to obtain different granularities and multilevel features.

On the contrary, some works suppress partial features to help model focus on small foreground objects. SDP [21] proposes a scale decoupling module to mask large object features and concentrate on small objects. Additionally, it introduces a sparse nonlocal attention mechanism and an adaptive anchor matching strategy to enhance small object detection performance. FSDet [22] also faces the extremely imbalanced foreground–background distribution. They design a class-aware context aggregation module to integrate the intraclass context information and suppress the background context.

The selection of more suitable anchors for aerial images is also explored. DEA-Net [23] proposes a novel training generator which interacts between an anchor-based unit (RoI) and an anchor-free unit in order to improve the quality of anchor. Facing misaligned issues in long-narrow and multiscale geospatial objects, an efficient feature aligned single-shot detector (ASSD) [24] proposes a pseudo anchor proposal module and a context-based feature alignment module. To address the variation problem in the scale and arbitrary orientations, HAA-Net [25] utilizes a hierarchical adaptive alignment network instead of relying on preset anchors with variable scales, angles, and aspect ratios.

Considering the redundant background problem, ClusDet [1] predicts object cluster regions to identify candidate clusters with higher object density compared to other regions using CPNet. This is followed by ScaleNet to unify the scales, and finally DetecNet is employed to obtain the final results. PENet [26] utilizes a coarse anchor-free detector to effectively predict the center point of the small object clusters, and a fine anchor-free detector to locate the precise positions of the small objects within these clusters. UFPMP-Det [2] introduces foreground region generation to predict subareas, which are then assembled into a unified mosaic to improve the foreground ratio. However, all methods above predict subareas of the input images first and detect objects in a second stage, hampering an end-to-end pipeline.

QueryDet [5] predicts the coarse locations of objects on high-level features and then computes the accurate detection results with sparse low-level features guided by those coarse positions. Sparse convolutions are employed to process these sparse features. However, this method requires special operations like sparse convolution, making it difficult to apply in an industry scenario. When using sparse convolution, all multiscale feature maps obtained by FPN are utilized for predicting objects. In contrast to QueryDet, which utilizes all feature maps, our method only sends L foreground areas (we set L to 2000 in our experiments) to the transformer detection head. Therefore, while sparse convolution is faster than transformer operations, its efficiency is still inferior to AODet due to the computational burden imposed by processing all multiscale feature maps.

C. Vision Transformers (ViTs) for Detection

With its successful application in the natural language processing (NLP) field, transformer has also been migrated to the visual field. ViT [27] is the first successful migration of the pure transformer architecture in the visual tasks. Detection transformer (DETR) [28] directly trains the model using bipartite matching loss, motivated by the idea that object detection can be regarded as a "set prediction" problem. Due to the design that one ground-truth box is only assigned to one predicted bounding box, the non maximum suppression (NMS) procedure can be removed. Conditional-DETR [29] and Anchor-DETR [30] add position information in the input query, making model much easier to convergence. However, these methods are designed for general object detection, which are always dense prediction. Therefore, they may not be suitable for aerial object detection considering the aforementioned problem.

Deformable DETR [31] selects a small set of keypoints around the target positions in feature maps, achieving better accuracy and faster convergence. Two-stage deformable DETR generates region proposals in the first stage, which are then fed into the decoder as object queries for further refinement. AODet is different from two-stage deformable DETR as follows. First, AODet utilizes foreground proposal network to obtain the feature of foreground proposals, which are directly sent to Transformer detection head to predict instances. While the first stage of deformable DETR predicts the class and bounding box of possible proposals. These predictions are computed separately by the class embed branch and the bbox embed branch and then sent into decoder as queries. Second, Transformer detection head of AODet consists of self-attention module and FFN and take foreground proposals as query, key, and value. However, the decoder of two-stage deformable DETR consists of self-attention module, cross attention module, and FFN. For cross attention, it takes foreground proposals as query and multiscale feature maps as key and value. Considering that selective foreground proposals are much fewer than all multiscale feature maps, AODet is much more efficient. Third, for aerial object detection, AODet is more suitable and simpler. Two-stage deformable DETR only exploit 300 queries in the second stage. When it comes to aerial object detection, it results in missing objects. The

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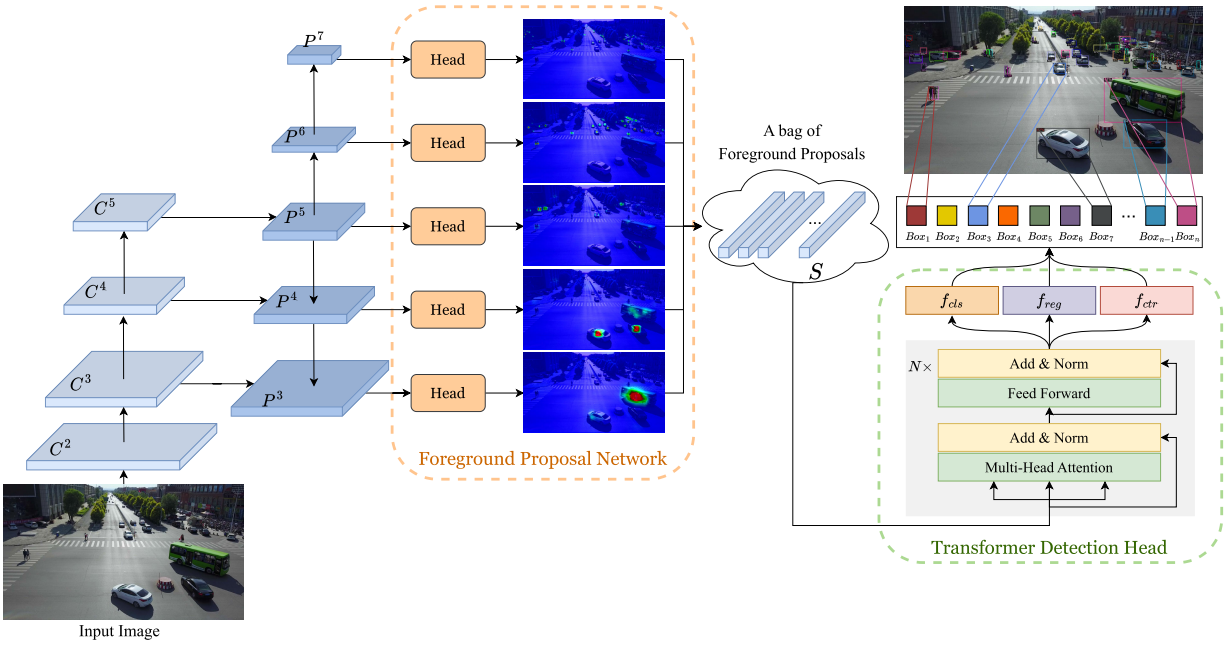


Fig. 2. Architecture of AODet. P_i denotes the multilevel feature maps, where $i = 3, 4, \ldots, 7$. "foreground proposal network" aims to predict a bag of foreground proposals, which are sent to "transformer detection head" to obtain the final detected boxes.

utilization of one-to-one assignment rather than many-to-one assignment aggravates this phenomenon. Consequently, two-stage deformable DETR misses amounts of objects while AODet can detect exactly.

III. Our Method

Traditional object detectors such as FCOS typically perform dense convolutions on the whole feature maps/input image, allocating equal computational budget to all the regions of an image. As discussed above, for aerial images, this simple strategy is not economical since large part of the image is easy background (e.g., sky) and can be discarded earlier. Here, we design a framework that can leverage this important observation. The network architecture of our method is illustrated in Fig. 2. An input image is first sent to backbone and FPN to extract multilevel feature maps. Foreground proposal network distinguishes the foreground regions from background regions of multilevel feature maps. Then, Transformer detection head applies the selected foreground regions and predicts the class and bounding box for the final object instances.

A. Backbone and FPN

The input image I first goes through a backbone network (e.g., ResNet [32]), which is followed by a FPN [12]. The outputs of FPN are multiple levels of feature maps with varying resolutions, denoted by $P_3, P_4, \ldots, P_7$ from high to low resolution, respectively.

B. Foreground Proposal Network

The goal of foreground proposal network is to predict the regions where the objects are more likely to appear. As shown in Fig. 3, foreground proposal network is composed of a classification branch and a regression branch. Each branch contains one convolutional layer (which has 256 output channels) with ReLU and group normalization [33]. These components are deliberately designed to be lightweight in order to save the computational overheads. Following FPN, they produce F^i_{cls} and F^i_{reg}, respectively, where $i = 3, 4, \ldots, 7$. These features are concatenated into $[F^i_{\text{cls}}, F^i_{\text{reg}}]$, which are then selected by probability map to obtain foreground proposal features. Probability map is computed by a convolutional layer with a sigmoid layer in F^i_{cls}, indicating the probability of being one of the foreground classes.

The selected foreground proposal features are determined as follows. For a i-layer feature vector at location (x, y), denoted by $F^i_{x,y}$, we obtain the score vector $p^i_{x,y}$ at the corresponding location on the probability map, which should be a C dimension vector. Since there are foreground objects, the corresponding feature vectors are then sent to the next stage. We utilize the maximum score in the score vector as the objectiveness score of the feature vector at location (x, y). Afterward, we select the L feature vectors from all $[F^i_{\text{cls}}, F^i_{\text{reg}}]$ layers, where $i = 3, 4, \ldots, 7$, with largest scores as the selective foreground proposal features $[S_{\text{cls}}, S_{\text{reg}}]$. To better integrate classification and regression features, a linear layer with LayerNorm is applied to obtain the final proposal features S. These final proposal features constitute the bag of features that the Transformer detection head will take as inputs.

1) Auxiliary Regression Loss: Additionally, as depicted in Fig. 3, in parallel with the local classification loss, we also incorporate an auxiliary regression loss attached to the feature maps F^i_{reg}. The auxiliary regression loss need another Conv-ReLU operations which has four output channels depicting the bounding box $t^i_{x,y} = (l, t, r, b)$ as FCOS. In our experiments, this auxiliary regression loss is only used during training, and we observed that it can greatly help the optimization and improve the performance in inference.

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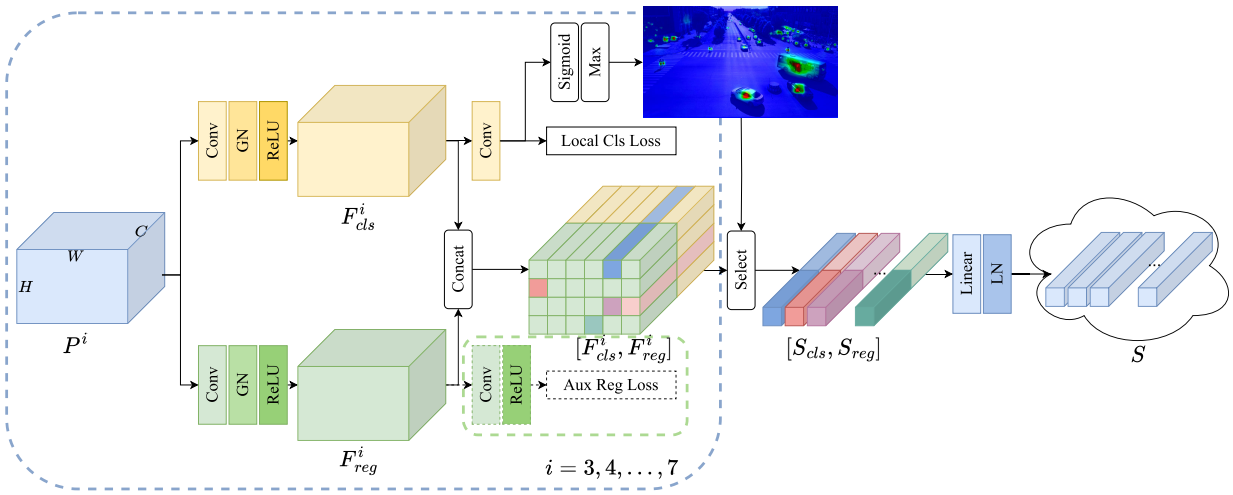


Fig. 3. Details of foreground proposal network. The foreground proposal network consists of classification branch and regression branch. Both branches contain one convolutional layer with ReLU and group normalization. F^i denotes feature map of layer i from FPN and S denotes the selected foreground proposals.

2) Dynamic Label Assignment: The goal of the Foreground proposal network is to choose the top L foreground proposals for the subsequent Transformer detection head. Consequently, it is crucial that these L proposals cover as many individual instances as possible, rather than being dominated by multiple proposals from the same instance. To achieve this goal, we impose a constraint during label assignment in training, limiting the maximum number of positive proposals per instance. This ensures that the selected L proposals can encompass a broader range of instances. Specifically, we first determine the labels in the same way as FCOS. Subsequently, for each instance, we choose K locations with the minimum classification and regression loss as the positive samples, while others are set as negative. Instead of using a fixed K here, following [34], we dynamically estimate K by summing up the largest Q (we set $Q = 50$ in our experiment) Intersection over Union (IoU) between the ground-truth and predicted boxes. Refer [34] for the details.

3) Difference Between Foreground Proposal Network and RPN: First, it is important to note that the regression branch of foreground proposal network is optional. While this auxiliary regression branch can help optimization during training and improve performance in inference, its absence does not detrimentally impact our overall framework. However, the regression branch of RPN is essential. RPN needs to predict bounding boxes (i.e., proposals) for following operations like RoIPooling [9] or RoIAlign [14]. Second, foreground proposal network utilizes dynamic label assignment in order to cover as many individual instances as possible rather than contain many proposals from the same instance. RPN follows traditional label assignment without such specialized design considerations.

C. Transformer Detection Head

In the Transformer detection head, the L feature vectors are first added with two types of positional encodings. First, as [28], we employ 2-D positional encodings to encode the spatial locations of these feature vectors on the original 2-D feature maps. Second, we incorporate learnable positional encodings to denote the FPN layer to which these feature vectors belong. Subsequently, these feature vectors are aggregated into a $L \times C$ matrix[2] and sent to the Transformer detection head. The Transformer detection head consists of $N = 6$ stacked self-attention and fully connected layers with $D = 2048$ channels. Following [7], we also use multihead attention, which allows the model to attend to different feature vectors in the inputs. The head number is set to $H = 8$.

The output of the transformer remains still a matrix with dimensions $L \times D$. We make use of three linear layers to map the $L \times D$ matrix into the final predictions, which include a classification branch to attain score vector p with dimension of class number for classification, a 4-D vector denoting the bounding-box's coordinates $t = (l, t, r, b)$ and a 1-D vector ctr to predict "center-ness" as in FCOS. Note that since we are aware of the original locations of these foreground proposals features, the final boxes can still be restored by using the features' location coordinates. If the location falls into any ground-truth box, it is considered as a positive sample, and the location's classification and regression targets are thus computed according to this ground-truth box. The classification target of each location is the class $\boldsymbol{c^*}$ of the corresponding ground-truth box. Otherwise, it is a negative sample $\boldsymbol{c^*} = \mathbf{0}$. The regression target (only available for each positive sample) is a 4-D vector $\boldsymbol{t^*} = (\boldsymbol{l^*}, \boldsymbol{t^*}, \boldsymbol{r^*}, \boldsymbol{b^*})$ to depict the target box for this location, where $\boldsymbol{l^*}$, $\boldsymbol{t^*}$, $\boldsymbol{r^*}$, and $\boldsymbol{b^*}$ denote the distances from the location to the four sides of the bounding box. The "center-ness" target $\boldsymbol{ctr^*}$ indicates how far away from an instance center, defined as FCOS.

Alternatively, one can adopt the bipartite matching as in DETR to directly predict the bounding-boxes' coordinates in the image. However, we empirically find it results in lower performance. We argue that using the locations of the feature vectors to restore the boxes can reduce the search space of the model and thus eases the learning difficulty.

[2]Note that we omit the batch dimension for clarification.

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D. Loss Functions

We define our training loss function as follows:

$$L_{\text{overall}} = L_{\text{fp}} + L_{\text{td}} \tag{1}$$

$$L_{\text{fp}}\left(p^i_{x,y}, t^i_{x,y}\right) = \frac{1}{N_{\text{fp_pos}}} \sum_{i}^{3,\ldots,7} \sum_{x,y} L_{\text{cls}}\left(p^i_{x,y}, \boldsymbol{c}^{i*}_{\boldsymbol{x,y}}\right) + \frac{1}{N_{\text{fp_pos}}} \sum_{i}^{3,\ldots,7} \sum_{x,y} L_{\text{reg}}\left(t^i_{x,y}, \boldsymbol{t}^{i*}_{\boldsymbol{x,y}}\right) \tag{2}$$

$$L_{\text{td}}\left(p_{x,y}, t_{x,y}\right) = \frac{1}{N_{\text{td_pos}}} \sum_{L} L_{\text{cls}}\left(p, \boldsymbol{c}^*\right) + \frac{1}{N_{\text{td_pos}}} \sum_{L} L_{\text{reg}}\left(t, \boldsymbol{t}^*\right) + \frac{1}{N_{\text{td_pos}}} \sum_{L} L_{\text{ctr}}\left(\text{ctr}, \boldsymbol{ctr}^*\right) \tag{3}$$

where L_{fp} is the loss function of foreground proposal network, and $N_{\text{fp_pos}}$ denotes the number of positive samples in foreground proposal network. The classification loss L_{cls} is computed with the focal loss [35] and regression loss L_{reg} is the IoU loss proposed in UnitBox [36]. $\boldsymbol{c}^{i*}_{\boldsymbol{x,y}}$ and $\boldsymbol{t}^{i*}_{\boldsymbol{x,y}}$ stand for the class and bounding boxes of the corresponding ground-truth box at location (x, y) in i-layer, respectively. L_{td} is similar to L_{fp} but the predictions of Transformer detection head. $N_{\text{td_pos}}$ denotes the number of positive samples in Transformer detection head, which is smaller than L. The centerness loss L_{ctr} is binary cross entropy (BCE) loss. The overall loss L_{overall} is sum of the foreground proposal network loss L_{fp} and Transformer detection head loss L_{td}.

E. Inference

For a given image, we first forward it through the network and obtain the prediction outputs of foreground proposal network. Subsequently, we select $L = 2000$ foreground proposal features, which are then sent into the Transformer detection head for predicting the final classification results and (l, t, r, b) of the bounding-boxes. The coordinates are decoded in the same way as FCOS thus the final bounding-box can be restored from the predicted 4-D regression vector and the location coordinates. Finally, NMS with IoU threshold being 0.6 is used to merge these predicted bounding-boxes.

IV. Experiments

We benchmark on common aerial dataset VisDrone and DOTA to demonstrate the superiority of our methods. Unless specified, all ablation experiments are conducted with VisDrone and ResNet-50 is used as backbone.

A. Datasets

VisDrone-Det is an object detection dataset with drone perspective. In Visdrone-Det, there are 6471 images in the training set, 548 images in the validation set, and 1580 images in the test-challenge set, all of which are labeled with 11 categories. This dataset is very challenging for object detection tasks due to the various scales of objects with the flying height and most of the objects are very small (less than 32 pixels). **DOTA** is a large-scale dataset for object detection in aerial images, where objects exhibits a wide variety of scales, orientations, and shapes. Here, we use DOTA v1.0 throughout this article. There are 1411 images in the training set, 458 images in the validation set, and 937 images in the test set, all of which are labeled with 15 categories. Note that in DOTA, there are two different tasks. One is oriented bounding boxes (OBBs) task and another is horizontal bounding boxes (HBBs) task. We focus on the HBB task in this article.

B. Implementation Details

Our approach is implemented based on PyTorch [4] and Detectron2 [43]. All models are trained on 8 NVIDIA A100 GPUs. For VisDrone, following QueryDet, we equally split one image into four nonoverlapping patches and process them independently during training. In inference, the whole image is used without splitting. We train the model for 50K iterations with batch size 16. The learning rate starts from 0.01 and decayed by a factor of 10 after 30 and 40K iterations, respectively. We apply data augmentation in our method, such as random flip and resize. Considering there are some blur occasions in VisDrone, we also use random blur and motion blur. Some images are obtained at night so that we add random contrast and random brightness as well. For DOTA, we follow FCOS and train with $1\times$ schedule and $3\times$ schedule.

C. Effectiveness of AODet

To verify the effectiveness of AODet, we first compare our method with general object detector FCOS which performs dense convolutions on the whole image/feature maps to show our proposed solution can attain better performance and runs faster by discarding the background regions earlier. When it comes to low foreground ratio in aerial images, AODet lead to much acceleration due to the fact that larger parts of background regions are discarded before Transformer detection head. As shown in Table I, when the foreground ratio is low (i.e., objects in images are more sparse), our method is $1.67\times$ faster than FCOS in VisDrone (5.0 versus 3.0 FPS). Moreover, it can be noted that our method can improve the detection AP (4.6 in VisDrone).

D. Ablation Study

1) Number of Foreground Proposals: Here, we investigate how the number of foreground proposals L affect the performance and inference speed. As shown in Table II, by changing L from 1000 to 1500, the AP and AP_{50} are improved by 0.6% and 1.4%, respectively, while the FPS is 5.4 and 5.2 FPS, respectively. When we further increase L to 2000, the FPS drop further with minor improvement in AP (0.3%). Increasing L from 2500 to 3000 brings no gain or loss in accuracy and FPS. Therefore, we choose $L = 2000$ in all other experiments. We also find that almost half of the foreground points are mainly related to other foreground points on the same layer, referring to Fig. 4.

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TABLE I

COMPARISONS OF FCOS AND AODET ON MS COCO AND VISDRONE. FG RATIO STANDS FOR FOREGROUND RATIO

Dataset	FG Ratio(%)	FCOS				AODet			
		AP	AP_{50}	AP_{75}	FPS	AP	AP_{50}	AP_{75}	FPS
COCO	44.0	38.7	57.5	41.6	16.2	40.1 (**1.4**+)	59.5	43.7	17.0 (**1.05**×)
VisDrone	9.7	31.7	52.5	33.0	3.0	36.3 (**4.6**+)	63.2	36.1	5.0 (**1.67**×)

TABLE II

ABLATION STUDY ON THE NUMBER OF FOREGROUND PROPOSALS (L)

L	AP	AP_{50}	AP_{75}	FPS
1000	35.4	61.2	35.1	**5.4**
1500	36.0	62.6	35.9	5.2
2000	36.3	63.2	36.1	5.0
2500	**36.5**	**63.5**	**36.3**	4.9
3000	**36.5**	63.2	**36.3**	4.9

TABLE III

EFFECTIVENESS OF DIFFERENT COMPONENTS IN FOREGROUND PROPOSAL NETWORK

Methods	AP	AP_{50}	AP_{75}
base	35.6	62.4	35.2
base+Aux Reg	35.9	62.4	35.7
base+Aux Reg+dynamic	**36.3**	**63.2**	**36.1**

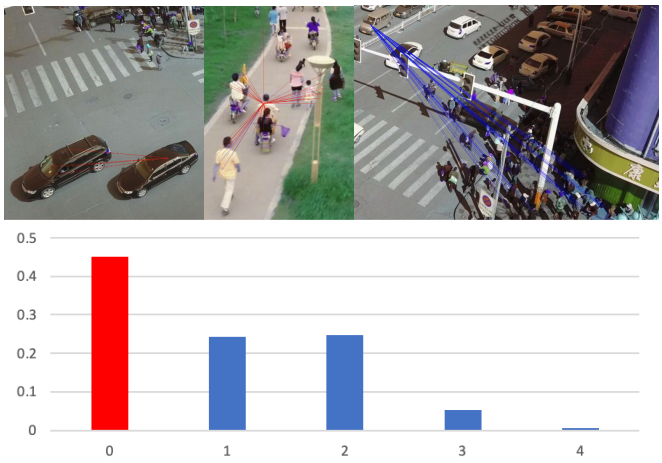


Fig. 4. Ratio of the relationship between foreground points between different FPN layer. 0 means that the two points are on the same layer, 1 means that the FPN of the two points is one layer different, and so on.

2) Effectiveness of Components in Foreground Proposal Network: As mentioned in "foreground proposal network," the auxiliary regression loss and dynamic label assignment strategy can improve the performance. We confirm it in this experiment. As shown in Table III, by using the auxiliary loss, the performance can be improved from 35.6 to 35.9 AP. Dynamic label assignment strategy further enhances the accuracy to 36.3 AP. As shown in Fig. 5, the average of standard deviations for all images is 1.82, which is smaller than 2.22, indicating that with the help of dynamic label assignment strategy, foreground proposals can assigned to ground truth boxes more equally.

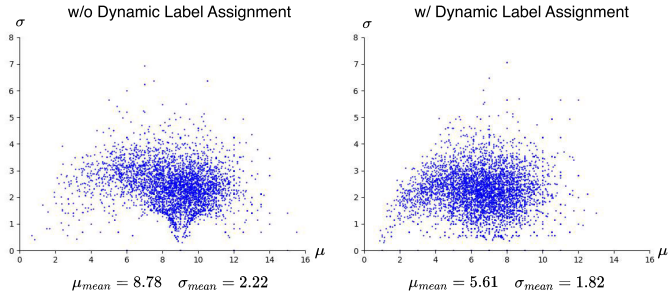


Fig. 5. Mean and standard deviations distribution of the number of foreground proposals assigned to each GT box in each image.

TABLE IV

IMPACT OF THE NUMBER OF STACKED SELF-ATTENTION LAYERS N AND THE DIMENSIONS OF THE HIDDEN LAYERS D IN TRANSFORMER DETECTION HEAD

N	D	AP	AP_{50}	AP_{75}	FPS
2	2048	35.1	61.7	34.9	**5.5**
4	2048	35.2	62.8	34.3	5.3
6	512	34.9	62.3	33.9	5.3
6	1024	36.2	63.0	**36.1**	5.1
6	2048	**36.3**	**63.2**	**36.1**	5.0

TABLE V

DETR-STYLE VERSUS FCOS-STYLE SUPERVISION. AS YOU CAN SEE, THE FCOS-STYLE SUPERVISION ACHIEVES MUCH BETTER PERFORMANCE

Supervision Type	AP	AP_{50}	AP_{75}
DETR-style	34.8	61.2	33.9
FCOS-style	**36.3**	**63.2**	**36.1**

TABLE VI

ABLATION STUDY ON THE MAJOR COMPONENTS OF AODET. "AUG.": USING TRAINING DATA AUGMENTATIONS. FGPROPOSALNET STANDS FOR FOREGROUND PROPOSAL NETWORK AND TRANSDETHEAD FOR TRANSFORMER DETECTION HEAD

Method	Aug.	AP	AP_{50}	AP_{75}	FPS
AODet	✓	**36.3**	**63.2**	**36.1**	**5.0**
AODet	×	33.3	59.9	32.0	**5.0**
- FGProposalNet	×	31.7	52.5	33.0	3.0
- TransDetHead	×	30.6	53.6	30.5	4.2

3) Design Choices of Transformer Detection Head: Here, we investigate the design details of the transformer detection head. As shown in Table IV, we vary the number of stacked self-attention layers N and the dimensions of the hidden layers D. As shown in the table, using $N = 6$ and $D = 2048$ achieves the best performance (36.3 AP).

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TABLE VII

COMPARISONS ON VISDRONE VALIDATION SET. † DENOTES MULTISCALE TESTING. ‡ DENOTES RESULT ON VISDRONE TEST-DEV SET. UFP/MP REFER TO THE UFP MODULE AND MULTIPROXY DETECTION NETWORK OF UFPMP-DET

Method	Backbone	AP	AP_{50}	AP_{75}	AR_1	AR_{10}	AR_{100}	AR_{500}	params	FLOPs	FPS
ClusDet [1]	ResNeXt-101	28.4	53.2	26.4	-	-	-	-	-	-	1.3
DMNet [37]	ResNeXt-101	29.4	49.3	30.6	-	-	-	-	-	-	1.7
RRNet [38]	ResNet-101	29.1	55.8	27.2	**1.0**	**8.5**	35.2	46.1	-	-	-
InterleavedCascade† [39]	ResNet-50	30.1	58.0	27.5	-	-	-	-	-	-	-
DREN [40]	ResNeXt-152	30.3	-	-	-	-	-	-	31M	-	-
QueryDet [5]	ResNet-50	28.3	48.1	28.8	0.5	6.0	36.5	39.4	40M	499G	4.2
CDMNet [41]	ResNeXt-101	31.9	52.9	33.2	-	-	-	-	-	-	-
FENet [20]	Triple-ResNext152	34.5	59.7	32.3	0.6	1.2	35.7	51.6	-	-	-
SDP [21]	CSPDarknet53	30.2	52.5	30.6	-	-	-	-	97M	-	-
UFPMP-Det [2]	ResNet-50	36.6	62.4	36.7	-	-	-	-	32M / 33M	215G / 228G	3.6
UFPMP-Det† [2]	ResNeXt-101	40.1	66.8	41.3	-	-	-	-	-	-	-
FEA-Swin‡ [42]	SwinT	33.7	-	-	-	-	-	-	121M	-	-
AODet	ResNet-50	36.3	63.2	36.1	0.5	6.1	42.5	53.6	42M	172G	5.0
AODet	ResNet-101	37.7	64.5	37.9	0.5	6.3	44.1	54.5	61M	249G	4.5
AODet	Swin-B	39.6	67.9	39.6	0.6	6.5	45.4	56.4	105M	413G	1.8
AODet†	ResNeXt-101	40.2	68.0	40.7	0.5	6.5	45.7	57.2	104M	412G	-
AODet†	Swin-B	**40.9**	**69.4**	**41.4**	0.5	6.4	**46.0**	**57.9**	105M	413G	-

TABLE VIII

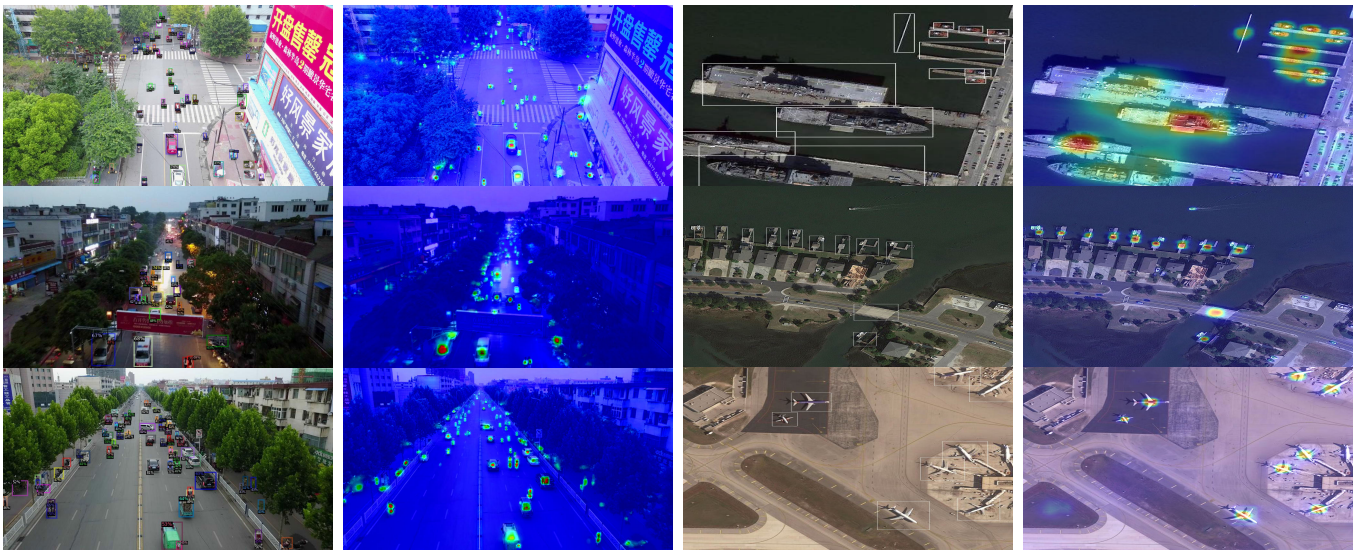
COMPARISONS WITH OTHER METHODS OF HBB TASK ON DOTA TEST-DEV SET. PL, BD, BR, GTF, SV, LV, SH, TC, BC, ST, SBF, RA, HA, SP, AND HC STANDS FOR PLANE, BASEBALL-DIAMOND, BRIDGE, GROUND-TRACK-FIELD, SMALL-VEHICLE, LARGE-VEHICLE, SHIP, TENNIS-COURT, BASKETBALL-COURT, STORAGE-TANK, SOCCER-BALL-FIELD, ROUNDABOUT, HARBOR, SWIMMING-POOL, AND HELICOPTER, RESPECTIVELY

Method	mAP	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
FR-H [44]	60.5	80.3	77.6	32.9	68.1	53.7	52.5	50.0	90.4	75.1	59.6	57.0	49.8	61.7	56.5	41.9
DAL [45]	64.3	89.7	70.8	23.9	50.4	50.3	72.4	78.3	90.6	61.9	79.2	64.2	63.2	61.5	67.4	39.5
SBL [46]	64.8	89.2	66.0	46.8	52.6	73.1	66.1	78.7	90.9	67.4	72.2	39.9	56.9	69.6	67.7	34.7
CFPANet [47]	65.3	88.7	71.3	49.0	52.1	48.6	73.4	61.1	90.5	57.6	84.1	66.6	62.7	73.3	57.6	42.8
FMSSD [48]	72.4	89.1	81.5	48.2	67.9	69.2	73.6	76.9	90.7	82.7	73.3	52.7	67.5	72.4	80.6	60.2
ICN [49]	72.5	90.0	77.7	53.4	73.3	73.5	65.0	78.2	90.8	79.1	84.8	57.2	62.1	73.5	70.2	58.1
EFR [50]	73.5	88.4	83.9	45.8	67.2	76.8	77.2	85.4	90.8	85.6	75.8	54.6	60.8	71.4	78.0	60.9
AF-EMS [51]	74.0	89.9	77.5	54.6	65.1	66.1	80.1	87.0	90.9	82.9	85.9	60.9	44.5	77.6	75.2	71.5
AOPDet [52]	74.3	90.2	77.9	47.5	74.3	73.6	78.7	**89.5**	90.9	66.4	85.7	57.1	**71.9**	75.6	72.8	60.9
SCRDet [53]	75.4	90.2	81.9	55.3	73.3	72.1	77.7	78.1	90.9	82.4	86.4	64.5	63.5	75.8	78.2	60.1
ASSD [24]	76.0	89.3	82.0	52.1	62.6	79.8	81.8	88.2	90.9	82.0	82.9	59.8	65.3	77.1	82.0	64.5
FSDet [22]	77.8	89.2	84.3	55.4	74.8	80.9	78.9	88.2	90.9	86.9	85.9	67.1	64.8	76.5	75.1	69.0
AMFFA-Net [54]	78.1	89.7	85.0	59.5	77.0	75.5	79.9	87.6	90.9	87.3	85.7	65.1	66.1	79.4	77.8	64.8
A^2RMNet [55]	78.5	89.8	83.4	60.1	73.5	79.3	83.1	87.9	90.9	87.0	87.4	60.7	69.1	79.9	79.7	65.2
NPMMR-Det [56]	78.9	**93.0**	83.5	61.4	70.8	75.1	85.1	90.5	**95.3**	86.3	**90.5**	67.8	71.8	80.3	71.4	61.4
DDBN [57]	79.3	89.8	85.3	61.2	**78.5**	79.7	84.0	88.4	90.9	84.3	87.1	65.2	68.8	**84.9**	81.8	59.5
SCRDet++ [58]	79.4	90.0	86.3	**65.0**	74.5	72.9	**84.2**	79.1	90.7	**87.4**	87.1	**72.1**	66.7	82.6	80.6	71.1
AODet	**79.6**	89.3	**86.6**	62.1	67.9	**82.6**	75.3	87.0	90.7	85.3	87.2	62.5	70.2	82.0	**82.9**	**82.7**

4) Supervision in Transformer Detection Head: Many transformer-based object detection methods such as DETR make use of the bipartite matching loss to directly supervise the global bounding-boxes coordinates (referred to as DETR-style) instead of supervising the coordinates relative to the features' locations (referred to as FCOS-style). It is also feasible to use the bipartite matching loss in our method. Therefore, we conduct experiments to investigate which one is better for our task. As shown in Table V, the FCOS-style is much better than DETR-style loss by 1.5 AP. We argue that the usage of locations can increase learning efficiency by reducing the search space.

5) Major Components of AODet: The major components of AODet are foreground proposal network and Transformer detection head. Removing foreground proposal network, our method cannot predict the foreground proposals and thus degenerate to dense prediction (as FCOS). Removing Transformer detection head, our method can usage sparse convolutions to convert foreground proposals to predicted objects (as QueryDet). The comparison are shown on Table VI.

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(a) (b) (c) (d)

Fig. 6. Visualization of the detection results and attention maps on VisDrone and DOTA. Results in VisDrone (a) and (b) show AODet locates objects accurately even when the instances are small and dense. AODet also performs well without sacrificing the performance of common and big objects, as shown in the first row of DOTA (c) and (d). (a) VisDrone Pred. (b) VisDrone Attn. (c) DOTA Pred. (d) DOTA Attn.

E. Comparisons With State-of-the-Art

We compare our results with other state-of-the-art methods on VisDrone in Table VII and DOTA in Table VIII. The accuracy reach 39.6 AP on VisDrone and we further use multiscale testing, which improves the performance to 40.9 AP, being better than previous methods. Although the performance of UFPMP-Det is just the same with AODet, it is much more complicated due to two-stage training schedule and clustering algorithm. The comparison on FPS, Params, and FLOPs shows that our proposed method is able to attain even better speed-accuracy trade-off than previous methods. We also compare AODet with other recent approaches on the DOTA test set in Table VIII, obtaining competitive results (79.6 AP).

F. Visualization

We visualize the detection results and the attention maps on VisDrone and DOTA on Fig. 6. From the attention map, we can find that AODet works as expected and is able to locate the foreground regions accurately, even when the location is extremely small or is occluded by the other staff (e.g., trees). When it comes to common or large instances, our model can also locate the objects accurately and thus get precious positions.

V. Conclusion

We have proposed a transformer-based aerial object detector AODet which leverages the observation that only a small portion of the pixels in images is foreground and the background is easy enough which can be discarded earlier. Extensive experiments on VisDrone and DOTA benchmarks show that our proposed method is able to attain better speed and accuracy than previous detectors specialized for aerial objects. Moreover, our method does not involve special operations or algorithms such as ROIPooling or sparse convolutions and thus can be highly efficient and easily implemented with modern deep learning frameworks. However, our method struggles when a high proportion of the image is occupied by foreground areas due to the presence of numerous small objects (i.e., parking lots or crowded streets). Selecting only a limited number of foreground areas may yield incomplete effective instance areas, resulting in missed objects. While opting for a more extensive extraction of foreground areas would increase the overall computational complexity and calculation time rapidly, owing to the square complexity of attention in the Transformer detection head. We will focus on improving this limitation in our future work.

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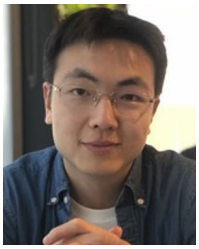
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Xiaoming Wang received the B.S. degree from the Northwestern Polytechnical University, Xi'an, China, in 2019, where she is currently pursuing the Ph.D. degree with the School of Computer Science.

Her research interests include computer vision, object detection, and instance segmentation.



Hao Chen received the Ph.D. degree in computer science from Australian Institute of Machine Learning, University of Adelaide, Adelaide, SA, Australia, in 2020.

He is a Research Professor with the College of Computer Science and the State Key Laboratory of CAD&CG, Zhejiang University, Hangzhou, China. Before joining ZJU, he was the Director Engineer at Huawei Noah's Ark Laboratory, Shanghai, China, leading the visual perception for self-driving project.



Xiangxiang Chu received the B.Sc. degree in electrical engineering from the SouthEast University, Nanjing, China in 2010, and the M.Sc. degree in electrical engineering from Tsinghua University, Beijing, China, in 2012.

He is currently a Senior Technical Manager at the Visual Intelligence Department, Meituan, Beijing, China. His current research interests include LLM pretraining, multimodal large models, reinforcement learning and generative models, foundation model design, NAS, model compression, self-supervised learning, 2-D perception, and 3-D detection.



Peng Wang received the B.E. degree in electrical engineering and automation and the Ph.D. degree in control science and engineering from the Beihang University, Beijing, China, in 2004 and 2011, respectively.

He is currently a Professor with the School of Computer Science at Northwestern Polytechnical University, Xi'an, China. His current research interests include computer vision, machine learning, and artificial intelligence.

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